

WEARABLE DEVICES: Essential Inputs for Design Engineers



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With the increase in usage of smartphones, opportunities for applications using mobile phones have also increased. This coupled with connectivity options such as Bluetooth and Near Field Communication and add-on devices to the phones have become very popular as these offer unique solutions. These devices can work as a simple radio-frequency identification reader to a sophisticated medical device. While the devices use conventional electronic circuitry, product packaging and usage are completely different. Right from the material used to the product shape and size have to be carefully selected to ensure safety of the wearer.

Most of these devices are small and irregular in shape. Engineering these needs special skills and manufacturing these needs special equipment and processes. Designers need to follow special design practices for wearable designs to be realised as successful products. There are very few books available for designers to design wearable products. This article tries to help design engineers to get an overall view of the design and manufacturing process of wearable devices.

Since wearable devices are normally made in large numbers and are miniature in size, use of contract manufacturing is recommended.

The focus is on the design process and what designers should look for when they design wearable devices to be made by contract manufacturers. While this list is not exhaustive, it gives a fairly good view of the complexities involved in the wearable device design.

Defining wearables

Wearable devices are no different from classic embedded systems that we are all familiar with. Block diagram

of a wearable device is shown in Fig. 1. It has following four blocks:

1. Power supply and management circuit
2. Core processor
3. Analogue interface [analogue front-end (AFE) covering sensor inputs and actuator outputs]
4. Communication interface (wireless and wireline)

What differentiates wearable devices from conventional embedded systems is the power management circuitry. Most wearable devices are battery-operated (either primary or chargeable) and need special circuitry to control power consumption.

Power management circuitry alone cannot optimise power consumption. It is the combination of the power management circuit and the device software in the system that results in optimal power consumption.

To a large extent we can say that, the processing power needed by wearable devices is very low, as most devices just collect data from sensors, store and send it to the central application for processing. The central application could be a smartphone or, in the case of the Internet of Things, cloud-mounted software. This is essentially done to ensure that the battery lasts long and data is processed in a central location by a more powerful processor.

Most integrated circuits (ICs) use very high-speed complementary metal-oxide semiconductor (CMOS) technology. In CMOS technology, power consumption of an IC goes up as the operating frequency of the processor goes up. So most wearable devices use low-speed controllers, and sections of the microcontrollers (MCUs) can be shut down or operated at a low speed when not in use.

We also need to remember that, some wearable devices that are used in medical applications (like insulin pumps or carry-on ECG probes) need to be designed in a controlled way, as these are approved by

